

Formula 1 Case

An F1 driver meets the end of the road...

- ▶ The F1 driver was killed in Nagoya: Sebastian Vettel found dead after a pre-race dancing night.
- ▶ His rental car was found burning next to the Port Bluebonnet Wildflower Garden in Nagoya.
- ▶ The burning is estimated to have occurred in the night of October 12nd 2019.
- ▶ The body of the victim was found in the trunk of the car.





One of the other F1 drivers may be the murderer

- ▶ At the night of the murder, drivers from all over the world were in Nagoya to have fun before the race.
- ▶ The competition between the teams is intense and the drivers always look for strong emotions.
- ▶ The excellent detectives of the Japanese police suspect, given the above, that one of the drivers may be the murderer.

Various items were found around the murder scene



Find the 5 W's

- ▶ Weapon (How)
- ▶ Where (Location of the murder)
- ▶ When (Time of the murder)
- ▶ Who (Person that committed the murder)
- ▶ Why (What is the motive)

What's expected from you?

- ▶ Present your case, including:
 - ▶ Your solution to the 5 W's
 - ▶ Your approach to finding this solution
 - ▶ Slides to support your story
- ▶ The winning team will be decided based on:
 - ▶ The technical approach to solving the case
 - ▶ The accuracy of the uncovered storyline
 - ▶ The final case presentation



Tips before starting



It's useful to keep in mind...

- ▶ To use project roles, divide the tasks, create a plan of approach
- ▶ That different steps in your investigation lead to new insights
 - ▶ For example, the where and when are evidently connected
- ▶ That Python, R and Excel can be used to solve the case. Other technologies are of course also welcome

Explaining the datasets

Explaining the datasets

Dataset 1 - Murder and weapon data

weapons.csv and murderdata.csv

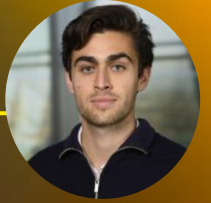


- ▶ In weapons.csv:
 - ▶ Contains the chemical samples of various murder weapons in previous crimes seen by the Japanese Police Department, including "Type" - the type of the murder weapon
- ▶ In murderdata.csv:
 - ▶ Two samples were found on the victim, containing different levels of the following chemical components RI, Al, Mn, Pb, Si, K, Cu, Ag, Fe

Explaining the datasets

Dataset 2 - GPS data

GPSTData.csv



- ▶ The burnt car was leased from a Japanese company that required telematics to be installed. This telematics unit amongst others was gathering the following data in the days before the murder:
 - ▶ Time
 - ▶ Latitude
 - ▶ Longitude

Explaining the datasets

Dataset 3 - Emergency call data

112Calls.csv



- ▶ The police department supplied us with the log of emergency calls in the days around the murder, containing:
 - ▶ Latitudes (N-degree)
 - ▶ Longitudes (E-degree)
 - ▶ Emergency type
 - ▶ Date
 - ▶ Time

Explaining the datasets

Dataset 4 - Historical call data

PhoneData.csv



- ▶ Information about phone history of the leading suspects (drivers) was found, containing:
 - ▶ Id (name)
 - ▶ Type -> call or text
 - ▶ Longitude
 - ▶ Latitude
 - ▶ Time (of the call/text)

Explaining the datasets

Dataset 5 - Tweets from drivers

Tweets.csv and nrc.xlsx



- ▶ In tweets.xlsx:
 - ▶ Provides information about tweets of each F1 driver
 - ▶ Contains 4 columns:
 - ▶ ID → identifier of tweet
 - ▶ Created_at → datetime of created tweet
 - ▶ Text → the actual tweet
 - ▶ TwitterName → user
- ▶ In nrc.xlsx:
 - ▶ Provides information about the sentiment of words and can be used for sentiment analysis.

Explaining the datasets

Dataset 6 - Fraud data

Fraud data sets



- ▶ In fraud.csv:
 - ▶ Contains the transaction records of the suspect in the past that were flagged as possible fraud which can be used for training purposes.
- ▶ In fraud_target.csv:
 - ▶ Contains the transaction records of the suspects of the day before the murder. This can be used to find the prime suspect(s).
- ▶ In suspects.xlsx:
 - ▶ Contains the bank accounts of the suspects

Get detecting!